

HTML `<meta charset>` and Character Encodings

The `<meta>` element provides metadata—information about the HTML document. The `charset` attribute specifies the character encoding used to interpret the document's text.

```
<meta charset="UTF-8">
```

This tells the browser that the document uses **UTF-8 encoding**.

Why character encoding is required

Computers store characters as numeric values. A character encoding defines how those numbers correspond to letters and symbols.

Correct encoding ensures that the browser displays:

- English letters
- Hindi and other regional languages
- Mathematical symbols
- Currency signs
- Emojis
- Special characters

Without the correct encoding, readable text may appear as corrupted characters, such as:

Ã©
â,¹
💎

Common character encodings

Encoding	Description	Example characters
UTF-8	Universal and recommended encoding for HTML5	English, हिंदी, ਪੰਜਾਬੀ, 中文, ₹, €
UTF-16	Unicode encoding using 16-bit code units	Supports most international characters
ISO-8859-1	Older Western European encoding	English, French, German and Spanish
Windows-1252	Microsoft's extended Western encoding	Western text and typographic symbols
US-ASCII	Basic English encoding	A–Z, a–z, 0–9 and basic punctuation
ISO-8859-15	Western European encoding with the euro sign	English and European languages
Shift_JIS	Older encoding used for Japanese text	Japanese characters
GB2312	Older encoding used for simplified Chinese	Simplified Chinese characters

Encoding	Description	Example characters
Big5	Traditional Chinese character encoding	Traditional Chinese characters

1. UTF-8

UTF-8 is the standard and recommended encoding for modern webpages.

```
<head>
  <meta charset="UTF-8">
  <title>UTF-8 Example</title>
</head>
```

It supports multilingual content:

```
<p>English: Welcome</p>
<p>Hindi: आपका स्वागत है</p>
<p>Punjabi: ਤੁਹਾਡਾ ਸਵਾਗਤ ਹੈ</p>
<p>Chinese: 欢迎</p>
<p>Currency: ₹ $ € £</p>
<p>Emoji: 😊 🎓 🖥️</p>
```

Advantages of UTF-8 include:

- Supports nearly all writing systems
- Compatible with ASCII
- Recommended by modern web standards
- Suitable for multilingual websites
- Widely supported by browsers and servers

2. UTF-16

UTF-16 is another Unicode encoding.

```
<meta charset="UTF-16">
```

However, it is generally not recommended for ordinary HTML files. UTF-8 provides better compatibility with web technologies and is usually more efficient for English-heavy content.

If UTF-16 is declared, the actual file must also be saved using UTF-16.

3. ISO-8859-1

ISO-8859-1, also called Latin-1, is an older encoding for Western European languages.

```
<meta charset="ISO-8859-1">
```

It supports characters such as:

A–Z

a-z
é
ñ
ö
ç

It does not support Hindi, Chinese, most emojis, or many modern symbols.

4. Windows-1252

Windows-1252 is an older Microsoft encoding used for Western languages.

```
<meta charset="Windows-1252">
```

It includes characters such as:

€
“
”
—
™

It was common on older Windows-based webpages, but UTF-8 should be preferred for new websites.

5. US-ASCII

ASCII supports basic English letters, numbers, and punctuation.

```
<meta charset="US-ASCII">
```

Examples of supported characters:

A-Z
a-z
0-9
! @ # \$ % & *

It does not directly support characters such as:

₹
é
ह
中
😄

Older syntax for character encoding

Older HTML documents may use:

```
<meta http-equiv="Content-Type"  
      content="text/html; charset=UTF-8">
```

In HTML5, the shorter form is preferred:

```
<meta charset="UTF-8">
```

Character encoding versus HTML character codes

Character encoding should not be confused with HTML character entities. Character entities are written inside HTML content to display reserved or special characters.

Character	Entity name	Numeric code	Hexadecimal code
<	<	<	<
>	>	>	>
&	&	&	&
"	"	"	"
'	'	'	'
Non-breaking space	 	 	
Copyright	©	©	©
Registered trademark	®	®	®
Trademark	™	™	™
Rupee	—	₹	₹
Euro	€	€	€
Pound	£	£	£
Yen	¥	¥	¥
Degree	°	°	°
Plus–minus	±	±	±
Multiplication	×	×	×
Division	÷	÷	÷
Heart	♥	♥	♥
Check mark	—	✓	✓

Example:

```
<p>5 &lt; 10</p>
<p>Tom &amp; Jerry</p>
<p>&copy; 2026 ABC University</p>
<p>Course fee: &#8377;5000</p>
<p>Temperature: 30&deg;C</p>
<p>Status: &#10003; Completed</p>
```

Browser output:

```
5 < 10
Tom & Jerry
© 2026 ABC University
Course fee: ₹5000
Temperature: 30°C
Status: ✓ Completed
```

Complete example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">
  <title>Character Encoding Example</title>
</head>
<body>

  <h1>UTF-8 Character Demonstration</h1>

  <p>English: Welcome</p>
  <p>Hindi: आपका स्वागत है</p>
  <p>Punjabi: ਤੁਹਾਡਾ ਸਵਾਗਤ ਹੈ</p>

  <p>Copyright: &copy; 2026</p>
  <p>Course fee: &#8377;5000</p>
  <p>Temperature: 28&deg;C</p>
  <p>Comparison: 10 &gt; 5</p>
  <p>Result: &#10003; Passed</p>
  <p>Emoji: 😊</p>

</body>
</html>
```

For modern HTML documents, use:

```
<meta charset="UTF-8">
```

The encoding declared in the document must match the encoding used when the file is saved.